Controller and system overview

ESP32-C5 DevKitC-1 v1.2 / PSRAM / physical pins

24V / USB-powered DevKit; socket spacing: 22.86 mm

J6

3V3	1	3V3
	2	EN
I2C0_SDA	3	GPIO2/LP_SDA
I2C0_SCL	4	GPIO3/LP_SCL
RTC_INT_N	5	GPIO0/RTC
UI_BUTTON_ADC	6	GPIO1/ADC1_CH0
CFG_ADC	7	GPIO6/ADC1_CH5
CAN2_TX	8	GPIO7/CAN2_TX
CAN2_RX	9	GPIO8/CAN2_RX
I2C1_SDA	10	GPIO9/SDA
I2C1_SCL	11	GPIO10/SCL
RF433_RX	12	GPIO26/RF
ZONE_ENABLE	13	GPIO25/ENABLE
EXP_5V	14	5V
GND	15	GND
	16	NC

C5 v1.2 PSRAM / J1 LEFT

J7

GND	1	GND
	2	GPIO11/U0TX
	3	GPIO12/U0RX
CAN_TX	4	GPIO24/CAN_TX
CAN_RX	5	GPIO23/CAN_RX
	6	NC/PSRAM
	7	GPIO27/RGB
UI_PIEZO_PWM	8	GPIO4/PIEZO
VIN24_ADC	9	GPIO5
	10	NC
	11	GPIO28/BOOT
GND	12	GND
	13	GPIO14/USB+
	14	GPIO13/USB-
GND	15	GND
	16	NC

C5 v1.2 PSRAM / J3 RIGHT

GPIO25 HIGH permits AC + DC commands

ZONE_ENABLE — R16 2.2k — ZONE_LED_A — D1 ZONE_ENABLE GREEN — GND

ON/ON:0V ON/OFF:1.06V OFF/ON:2.13V OFF/OFF:3.30V

AC / normally-open relay

ZONE_ENABLE — R15 10k — GND

D3 1N4148W

EXP_5V — K1 (A1 / A2 / 13 / 14) HF46F/5-HS1 — EACRET / AC2_IN

AC_RELAY_LOW

AC_RELAY_GATE — R17 1k — ZONE_ENABLE

Q1 AO3400A (1 / 2 / 3) — GND

AC total: 2 Arms (J8 + J9). External fuse before J15/J12.

TP2 — GND

EXP_5V total budget: 210 mA (coil + DC bias + all expanders).

Budgets are operating limits, not hardware current trips.

J25 — POWER_0V — 1 — 24V – IN

COM and POWER_0V: separate PCB wires, common external PSU negative.

Configuration / 2-way DIP / one ADC

SW1 206-2ST

CFG_SW1_N — 1 — 4 — GND

CFG_SW2_N — 2 — 3

DIP: GPIO6 ADC; buttons: GPIO1 ADC; GPIO5 24V ADC.

3V3 — R5 1k — CFG_SW1_N — R27 10k — CFG_ADC

3V3 — R6 1k — CFG_SW2_N — R28 20k — GND

C12 100nF — GND

3V3 — R8 10k — RTC_INT_N

DC / COM gate drive

EXP_5V — R19 10k — DC_OPTO_C

ZONE_ENABLE — R18 330 — DC_LED_A — U1 LTV-817S-TA1-C (1 / 2 / 3 / 4) — GND

DC_GATE — DC_ZONE_RETURN

Q2 AO3400A (1 / 2 / 3)

R20 47k — D2 BZX84C8V2 — DC_DRIVER_COM

DC command budget: 0.5 A total; verify VGS/VDS under load.

External COM reference required; verify OFF paths.

2 / UI + sensors

3 / Expansion

4 / Reference

5 / 24V power

6 / CAN FD 1

File: ulva_can.kicad_sch

7 / CAN FD 2

8 / 24V monitor

Functional schematic / 24V field control

Ulva

Sheet: /

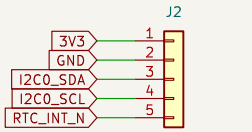
File: ulva_controller.kicad_sch

Title: Ulva / MCU and output control

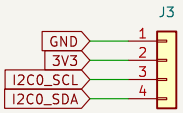
Size: A4	Date: 2026-10-13	**Rev: v0.42**
KiCad E.D.A. 9.0.9		Id: 1/8

UI, RTC and sensor interfaces

I2C0 display / RTC

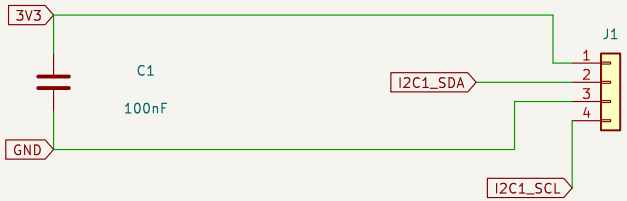


DS3231_RTC_SOCKET



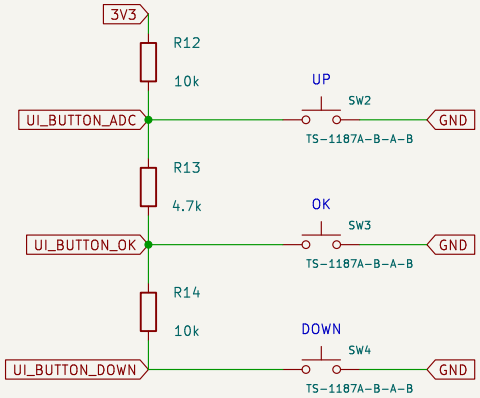
OLED 0.96 SSD1306 I2C

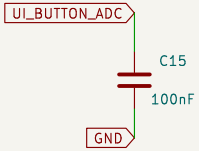
I2C1 (HP) / DHT20



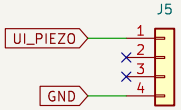
DHT20_LP_I2C

UP / OK / DOWN: GPIO1 ADC1_CH0



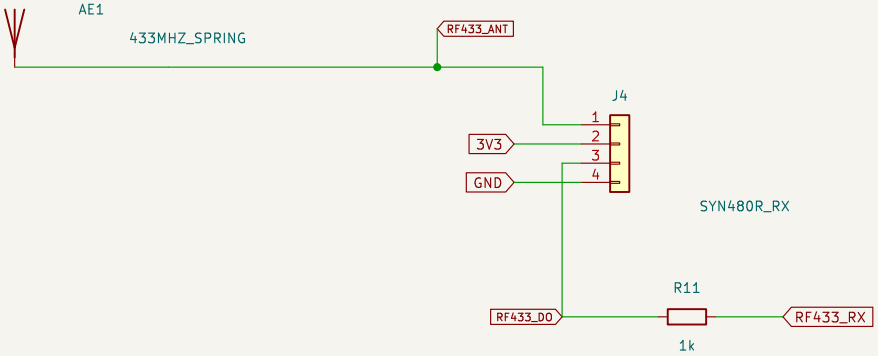


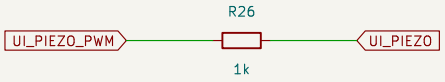
Passive piezo / dedicated GPIO4



PASSIVE PIEZO 1+ / 4−

GPIO26 / RF433 receiver: fitted; independent of RTC





1=red / 4=black; 2,3=NC. Male 2.54mm header.

Passive piezo only; no coil speaker or active module.

Independent PWM / RGB; verify loudness on hardware.

RTC INT: GPIO0 / deep-sleep wake; R8 10k pull-up on sheet1.

Functional schematic / 24V field control

Ulva

Sheet: /2 / UI + sensors/
File: ulva_ui.kicad_sch

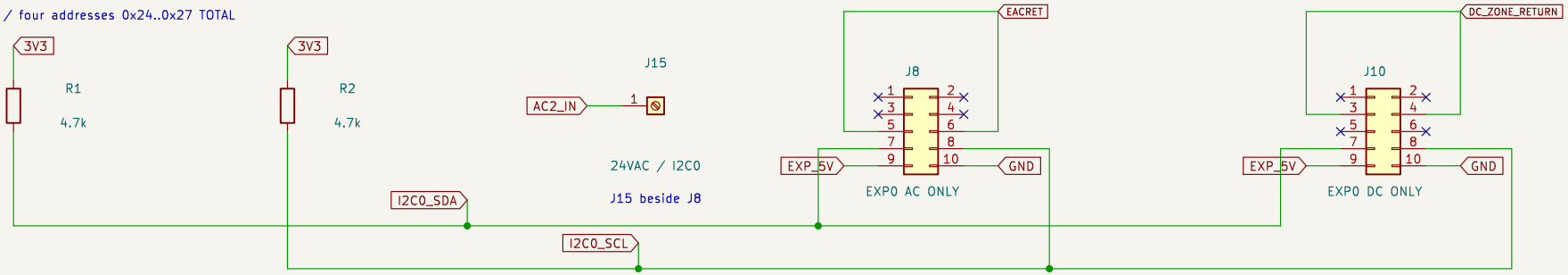
Title: Ulva / UI and RTC

Size: A4	Date: 2026-10-13	**Rev: v0.42**
KiCad E.D.A. 9.0.9		Id: 2/8

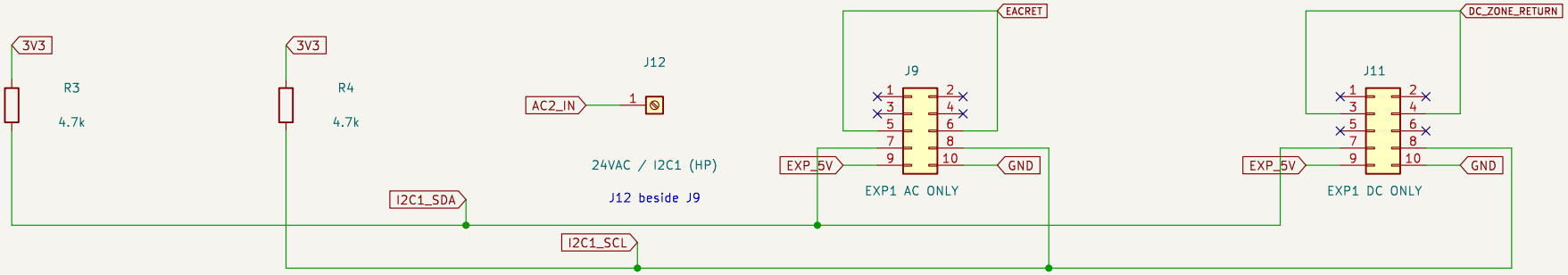
Expander buses, AC terminals and mounting

J12-J17: AC2_IN. J18/J19/J20-J23: six DRIVER COM inputs. J25: POWER_0V supply return.

I2C0 / EXP0: J8 AC + J10 DC / four addresses 0x24..0x27 TOTAL



I2C1 (HP) / EXP1: J9 AC + J11 DC / four addresses 0x24..0x27 TOTAL

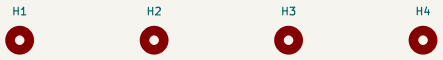


SDA/SCL: 3.3V pull-ups. EVIN (pin 9): EXP_5V. Pin 10: GND.

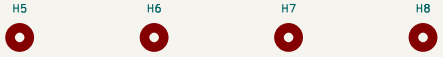
EACRET: AC load return (pins 5/6). DC_ZONE_RETURN: command return (pins 3/4).

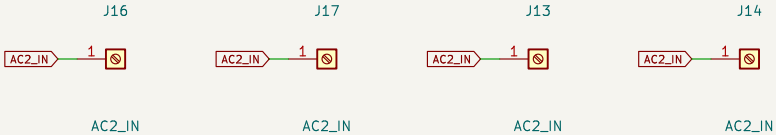
POWER_0V is bonded to GND by NT1. DRIVER COM is separate on PCB; external common reference required.

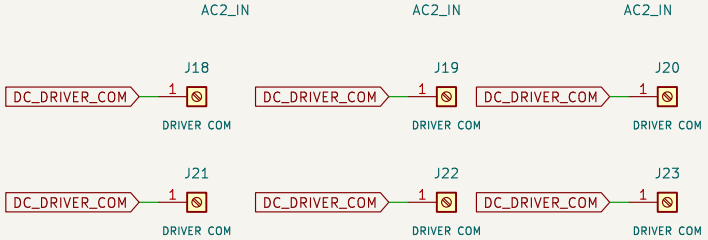
H1-H4 / M3 non-plated mounting holes



H5-H8 / OLED M2 slots, nylon hardware







Functional schematic / 24V field control

Ulva

Sheet: /3 / Expansion/

File: ulva_interfaces.kicad_sch

Title: Ulva / I2C and mounting

Size: A4	Date: 2026-10-13	**Rev: v0.42**
KiCad E.D.A. 9.0.9		Id: 3/8

External interface notes

External modules are separate hardware; they are not included in the carrier BOM or PCB.

AC0/J8 and AC1/J9 share one isolated 24VAC source and one K1 return switch.

J12–J17 are parallel AC inputs. Select one input and fit the required external inline fuse.

EACRET is distinct from logic GND. See docs/hardware.md for the complete IDC pin map.

DC0/J10 and DC1/J11 share DC_ZONE_RETURN. Q2 switches this return to DC_DRIVER_COM.

J18–J23 driver COM terminals and J25 POWER_0V are separate on this PCB.

The U1/Q2 gate drive requires driver COM and POWER_0V to share the external PSU negative.

External expanders can provide a return path around an open Q2; actual driver release needs testing.

Read docs/hardware.md before using the common–return switch to stop a driver.

Upstream OpenSprinkler Expander schematics and their license:

https://github.com/OpenSprinkler/OpenSprinkler-Hardware/tree/master/Expander/3.0

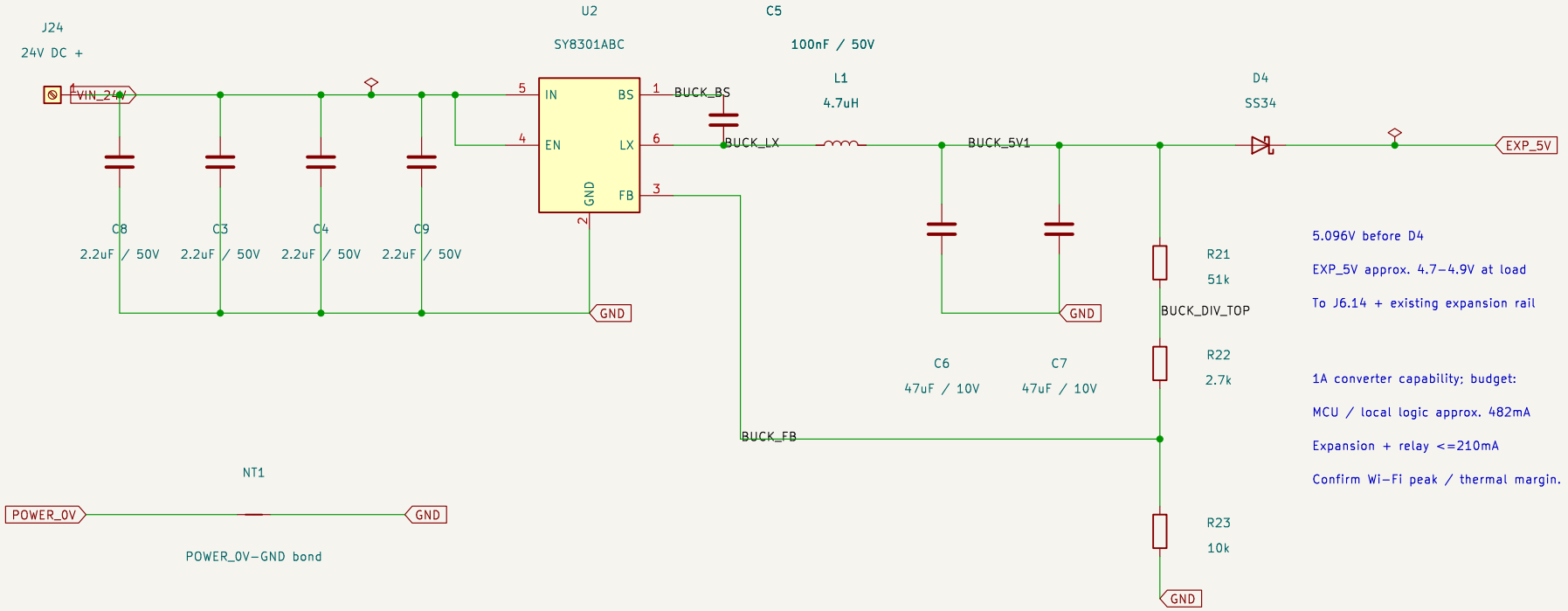
https://github.com/OpenSprinkler/OpenSprinkler-Hardware/blob/master/LICENSE.txt

See docs/hardware.md for pin assignments, connections and operating limits.

Ulva	
Sheet: /4 / Reference/	
File: ulva_reference.kicad_sch	
Title: Ulva / External interface notes	
Size: A4	Date: 2026-10-13
KiCad E.D.A. 9.0.9	Id: 4/8
	Rev: v0.42

24V DC input / DevKit + controller power

J24: +24V. J25: POWER_0V supply negative. DRIVER COM uses separate terminals; motor power stays external.



5.096V before D4

EXP_5V approx. 4.7–4.9V at load

To J6.14 + existing expansion rail

1A converter capability; budget:

MCU / local logic approx. 482mA

Expansion + relay <=210mA

Confirm Wi-Fi peak / thermal margin.

Existing J25 / 24V negative

NT1 is a copper connection, no part to buy.

DC_ZONE_RETURN remains switched by Q2.

Common GND changes external OFF-current paths.

Verify actual driver command release with the expander.

Assembly / PCB placement

U2 + L1: hand solder. Other added SMT: Basic.

U2 / L1 / capacitors: right of ESP32; J24: TOP-RIGHT.

Keep CIN-IN-GND and LX loops short; keep FB away from LX.

USB: D4 blocks USB -> buck. Official DevKit D1/D7 block VBUS backfeed; verify fitted board revision.

EN follows VIN, independent of ZONE_ENABLE. Controller remains powered when command enable is LOW.

PCB routed; power/USB/load and thermal tests remain.

Separate power return and DRIVER COM

Ulva

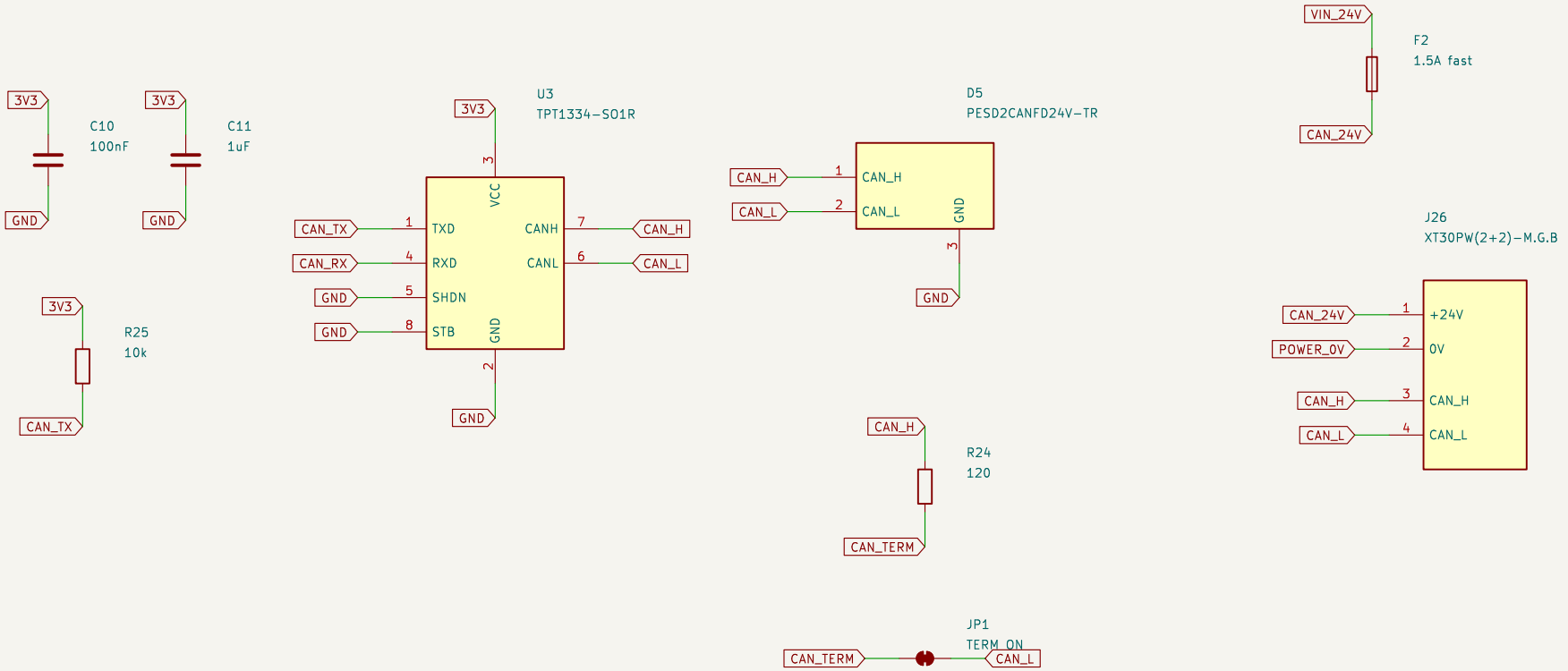
Sheet: /5 / 24V power/
File: ulva_power.kicad_sch

Title: Ulva / 24V to 5V logic supply

Size: A4	Date: 2026-10-13	**Rev: v0.42**
KiCad E.D.A. 9.0.9		Id: 5/8

CAN FD 1 / independent bus / ESP32-C5 first native controller

TX=GPIO24 (J7.4), RX=GPIO23 (J7.5). SHDN/STB fixed LOW; RGB and PSRAM retained.



JP1 closed: 120ohm across H/L. Endpoints only; no termination on intermediate motion extenders.

J24 +24V -> F2 -> J26.1. J26.2 returns directly to J25/POWER_0V; NT1 bonds logic GND.

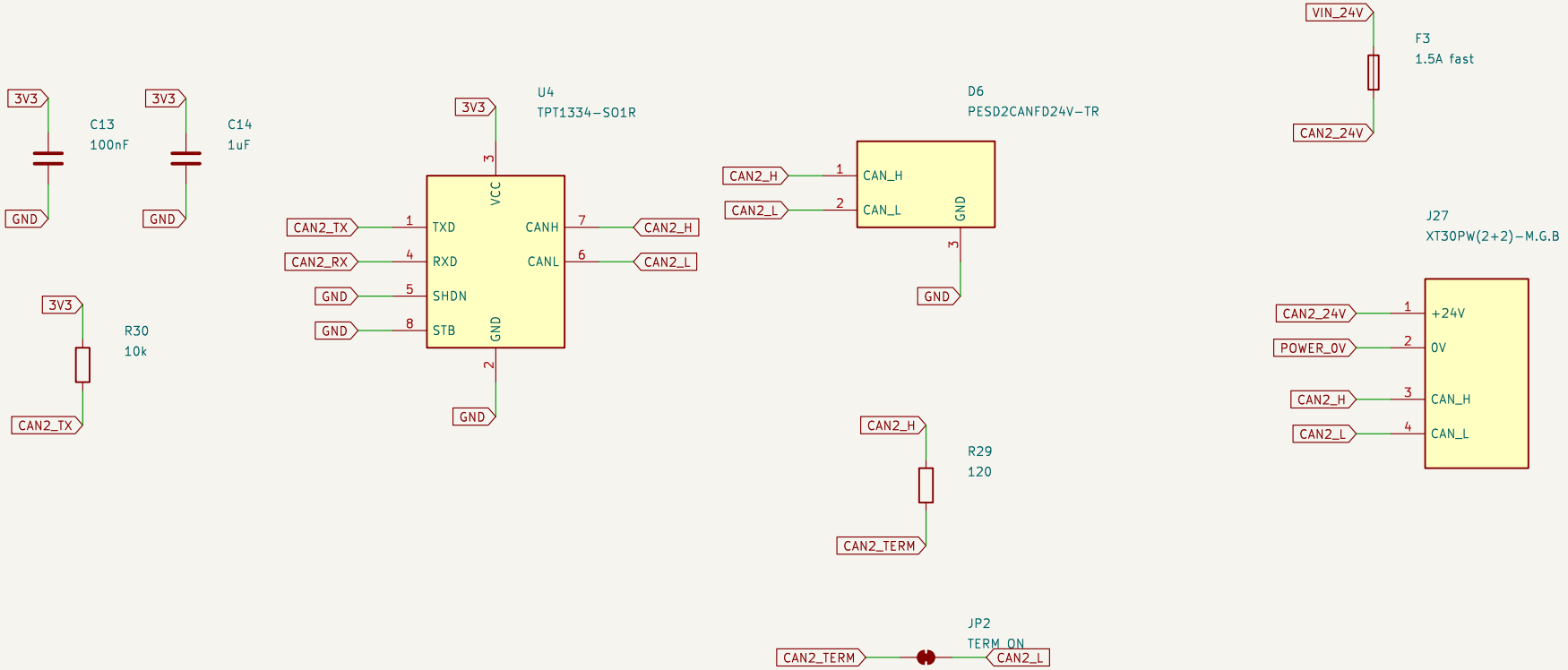
Design assumption: both CAN ports combined <=1A total, <=40C. Separate motor power. Fuse does not block reverse feed.

One source for the chain. Firmware, cable inrush, ESD, hot-plug and physical connector fit remain untested.

Ulva		
Sheet: /6 / CAN FD 1/		
File: ulva_can.kicad_sch		
Title: Ulva / CAN FD channel 1 + fused 24V power		
Size: A4	Date: 2026-10-13	**Rev: v0.42**
KiCad E.D.A. 9.0.9		Id: 6/8

CAN FD 2 / independent bus / ESP32-C5 second native controller

TX=GPIO7 (J6.8), RX=GPIO8 (J6.9). SHDN/STB fixed LOW; RGB and PSRAM retained.



JP2 closed: 120ohm across H/L. Endpoints only; no termination on intermediate motion extenders.

J24 +24V -> F3 -> J27.1. J27.2 returns directly to J25/POWER_0V; NT1 bonds logic GND.

Design assumption: both CAN ports combined <=1A total, <=40C. Separate motor power. Fuse does not block reverse feed.

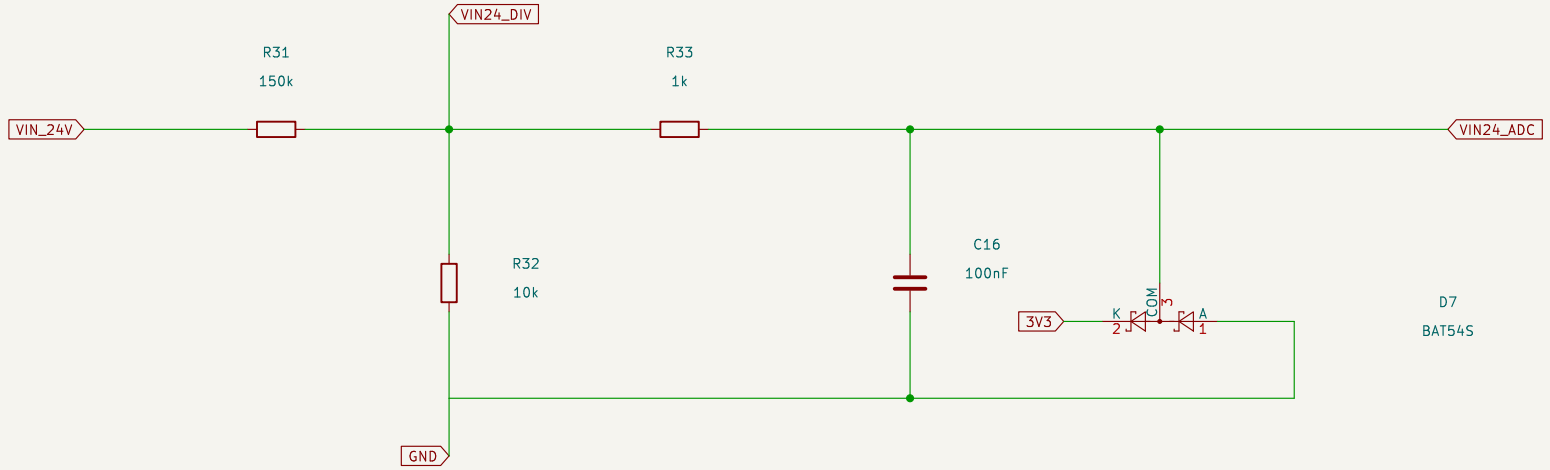
One source for the chain. Firmware, cable inrush, ESD, hot-plug and physical connector fit remain untested.

Independent H/L from CAN1. Shared GND / 3V3 / 24V source; no galvanic isolation.

Ulva	
Sheet: /7 / CAN FD 2/	
File: ulva_can2.kicad_sch	
Title: Ulva / CAN FD channel 2 + fused 24V power	
Size: A4	Date: 2026-10-13
KiCad E.D.A. 9.0.9	Id: 7/8
	Rev: v0.42

24V input monitor / GPIO5 ADC1_CH4 / J7.9

VIN_24V is sensed against logic GND. Existing regulator and ZONE_ENABLE are independent.



Nominal: 24V -> 1.500V, 36V -> 2.250V; multiply calibrated ADC voltage by16.

Rth=150k||10k+1k=10.375k; C16=100nF; tau=1.0375ms, fc=153.4Hz.

D7: pin1 GND -> diode -> pin3 ADC -> diode -> pin2 3V3 (manufacturer verified).

Rail clamp is not power-off isolation. If24V remains while3V3 is off, upper diode can backfeed3V3.

Calibration, leakage/temperature, actual ADC noise and startup/shutdown tests remain pending.

GPIO5 must remain an ADC input; no output drive or internal pulls. Shared ADC1 handle with buttons/DIP.

Four Basic passives + HJC C7420333 Preferred Extended. Monitor does not increase board input rating.

Ulva		
Sheet: /8 / 24V monitor/		
File: ulva_monitor.kicad_sch		
Title: Ulva / 24V ADC monitor		
Size: A4	Date: 2026-10-13	**Rev: v0.42**
KiCad E.D.A. 9.0.9		Id: 8/8